turgor

Illinois analysis + code

See Illinois soybean drought losses before harvest

Turgor turns soybean physiology (when drought and heat hit flowering and seed fill) into a weekly county stress map and expected loss, from satellite and weather data only. Tested on every Illinois county since 2001.

Why

Lenders, crop insurers and grain buyers learn the size of a soybean shortfall late. In Illinois the loss below trend averaged **US$ 23.5/acre a year** (2001–2025), **US$ 57/acre** in a bad year and **US$ 104/acre in 2012** (at US$ 12.55/bu). July–August rainfall is the usual weather signal; it can miss stress at seed fill.

What

Weekly, June to October: crop stage from MODIS NDVI (corn–soybean pixels, USDA Cropland Data Layer), water deficit, heat above 30 °C and their interaction from NASA POWER, weighted 3× at flowering and seed fill. Fixed physiology-based weights, not fitted to the yields they are tested on.

Illinois evidence (102 counties, 2001–2023 backtest; 2,194 county-seasons)

	Turgor	Rainfall Jul–Aug	Palmer Aug	NDVI only
AUC, loss > 10%	**0.81**	0.75	0.69	0.68
Share of losses missed (basis risk)	**0.47**	0.57	0.54	0.72
Statewide-year correlation	**0.60**	0.44	0.34	0.44

Honest limits: the gain over rainfall is not statistically proven in Illinois (better in 78–85% of resampled season sets), and on the 2024–25 holdout (17 losses) rainfall did better. **2003:** July–August rain looked normal, the index ranked the season 78th percentile of stress; yields ended −17%. **15 July 2012:** 99 of 102 counties rated severe; they ended −17%. **Timing:** by 1 September the index beats the end-of-season skill of the rainfall index, 4–6 weeks before harvest.

For whom

Reinsurers and parametric providers; lenders and crop insurers watching a soybean book in season; grain buyers forecasting supply. Pricing to test: per portfolio per year, or US$ 0.25–2 per acre per season.

Looking for

Conversations with lenders, insurers and grain buyers about how they track crop risk today. **Collaborators:** remote sensing and data science; agricultural finance and insurance. **Advisors:** crop insurance, farm credit, crop modelling.

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